How effective is fluid simulation in finding the power output of water turbines?

Mathematics

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Introduction

How effective is using computational fluid dynamics to test the power output of water turbines? This paper will explore how fluid simulation works and its application to water turbines. The complexity and what seems like random movement of fluids are very interesting. The study of these movements is crucial to many aspects of life, as fluids take up the majority of the world around us and have many helpful utilities to living. From subjects of engineering, physics, chemistry, math, and even art, fluids contribute to many questions on how and why the world works. As technology has advanced, equations have been found that can predict and display movements and interactions that fluids have with the environment, which has opened up many possibilities when utilizing fluids. This leads to using these equations to simulate scenarios involving fluids computationally. This is called computational fluid dynamics (CFD).

Hydropower is a very important resource when it comes to producing electricity in a clean and beneficial method. As hydropower uses water to create the energy in making electricity, it is a renewable source of electricity that helps sway away from the environmentally destructive methods of using fossil fuels. Along with generating electricity, hydropower also helps with flood control, irrigation, and with clean drinking water (EERE). When it comes to such a useful tool, it's important to maximize the benefits that come from it. This can be done by using the most efficient design for water turbines so that the most electricity possible can be produced. As technology advances, more accurate and efficient ways of creating and testing designs become available.

Water Turbines

To start, an understanding of what a water turbine is and how it works is needed. A water turbine is a piece of equipment that consists of runners that use the movement of water to turn shafts within an electric generator. From this, energy is created from a generator, converting mechanical energy into electrical energy.

Many design factors go into creating a water turbine and how those factors affect the power output of the turbine. These include the head, specific speed, rotational speed, cavitation, and the turbine shaft (theconstructor). The maximum efficiency of a water turbine can vary depending on the location that the turbine is located. This is due to space being a limiting factor to the head of a turbine, which is the difference of water levels between the hydro intake and the hydro discharge point. The efficiency of the turbine increases as the size of the head increases due to higher head sizes creating more water pressure to go through the turbine. In addition, as the size of the head increases, the cost and torque decreases, therefore a higher energy output (renewablesfirst). With this in mind, different head sizes will need different turbine designs to optimize for efficiency.



Figure 1. Voith, The distance between the different levels of water is the head.

There are two different categories of hydraulic turbines that define how the turbine uses the energy from the water (Linquip Team). One category is reaction turbines, which run on a continuous source of water. The shape of the runners on these turbines is important because they are used to create pressure as the water moves across the runner. The amount of mechanical energy output is determined by the change in pressure across the runners. Unlike reaction turbines, impulse turbines reduce the kinetic energy of water rather than the pressure energy. Impulse turbines use water jets to create torque against the runners that cause them to rotate. Since the designs use different kinds of energy to output power, they are calculated differently.

Fluid Simulation

Fluid simulation is a way to represent fluids computationally in a realistic manner. Fluid simulation can be done through various methods that all solve the fundamental equations of fluid dynamics. Fluids are made up of an infinite number of particles in real life which isn't practical to be done by computers, therefore calls for an alternate solution. The method showcased in this paper is the smoothed particle hydrodynamics method (SPH). SPH represents fluids using a finite number of particles in which each particle contains the properties such as velocity, density, pressure, etc (Ma).

In order to simulate a fluid, the Navier-Stokes equation is key for making it happen. These equations are what describe the flow of incompressible fluids (Müller et al). To do this computationally, a method has to be used to solve the equation using a finite number of particles. Here is where the SPH comes into play as it uses a constant number of particles within the system so the equation can be simplified.

$$\nabla \cdot u = 0$$

This is one of the Navier-Stoke equations, and the purpose of this is to represent the conservation of mass of a fluid. Within this equation, ∇ is the divergence for the velocity vector field of u. A vector field for this situation uses a vector for every particle of the liquid in the system and represents its direction. The divergence of the vector field is what determines how much of a point in the field is used as a source of the fluid. The divergence of fluids must equal 0 since a source of a fluid cannot be created or destroyed due to the conservation of mass, hence the equation (COMSOL). The other Navier-Stoke equation is as shown:

$$\rho(\frac{\delta v}{\delta t} + v \cdot \nabla v) = -\nabla p + \rho g + \mu\nabla^2 v$$

This equation is based on Newton's 2nd law, $F = ma$, in which the values are replaced using the forces put into liquid particles. In the case of the SPH method, the particles move with the fluid which means that the substantial derivative of the velocity field, $(\frac{\delta v}{\delta t} + v \cdot \nabla v)$, is the time derivative of the velocity of the particles, $\frac{Dv}{Dt}$, and therefore can be replaced in the equation. This leaves three force density fields in the equation that needs to be accounted for and implemented into the SPH for it to work. These density fields include pressure ($-\nabla p$), gravity (ρg), and viscosity ($\mu\nabla^2 v$). In the simplified equation, $\rho(\frac{Dv}{Dt})$, represents the change in momentum of the particles in the system, that being the sum of all the force density fields. Each field has an equation and computational problems that the SPH method works to solve.

The SPH method calculates the values and behaviors of particles by using the values of those around it. This is done using a smoothing kernel which uses the weighted average of values from neighboring particles in order to approximate the value for a particular particle (Hernquist and Katz).

This approximation of particle i is done using the equation, $f(A_i) = \sum_{j=1}^{N} \frac{m_j}{\rho_j} w_{ij} f(A_j)$. In this equation, A_j represents the position neighboring particles, and N represents the number of neighboring numbers within a radius of A_i. This radius is called the smoothing length (h) and has a big impact on the accuracy and stability of SPH (Zeng et al). The mass and density of particle j also contribute to the position of particle i ($\frac{m_j}{\rho_j}$). The weight value (w_{ij}) is determined by the use of kernel function.

The kernel function used in SPH utilizes the distance between particle i and particle j to determine the weight value, in which the closer the particles are to each other, the higher the weight will be. Therefore, the closer the neighboring particles are to the particle being calculated, the more of an impact that it will have on the position of that particle. This function also includes the smoothing length, giving the following equation.

$$w_{ij} = W(|A_i - A_j|, h)$$

There are a couple of kernel functions that are used when using the SPH method and depending on the function chosen can lead to a more accurate and stable model of fluids as it determines the pattern to interpolate. Kernel functions have certain properties that define them as such. One property that kernel functions provide is that it is always normalized, which allows for the approximation of constants exactly (Yang et al).

A commonly used kernel function for SPH method is the Gaussian kernel which is very practical due to its good stability and accuracy, though is inefficient computationally.

$$W(|A_i - A_j|, h) = \frac{1}{(\sqrt{\pi}h)^D} e^{-\frac{r^2}{h^2}}$$

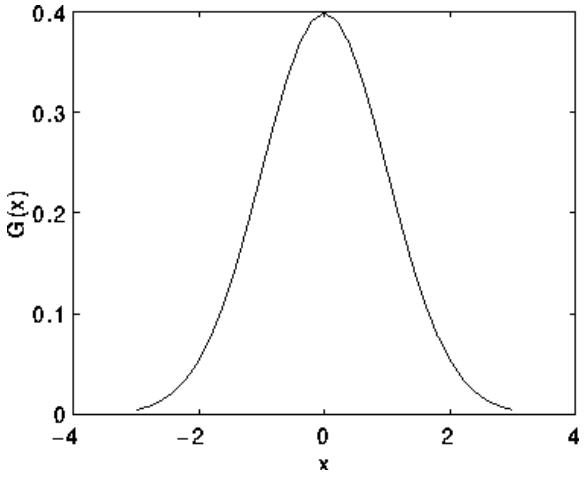


Figure 2. Fisher et al, Gaussian kernel curve.

D in this scenario equals 3, as it represents the number of dimensions being used. Since each particle of fluid has its own velocity, this means that it has a vector attribute to it. What the kernel function does is take the dot product of the vectors and uses the gaussian function to find a similarity value of the points (Johnson). This value is then able to give the distance between particles a new value that is in relation to the neighboring particles. This concept is the key behind the SPH method as it is a major determinant of the outcome of the simulation. With this, it is important to select the appropriate kernel function for the scenario being simulated as it has direct correlation to how accurate and consistent the particles are approximated.

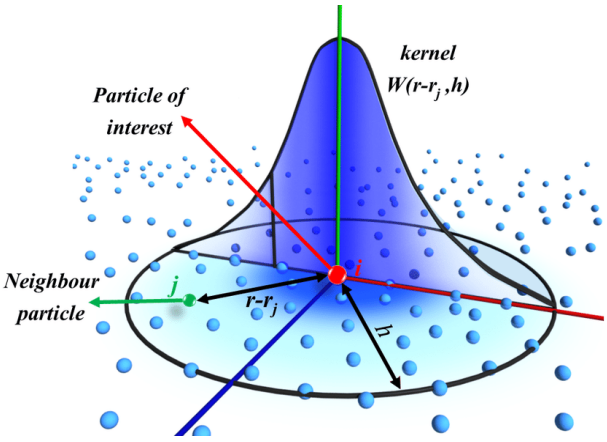


Figure 3. Giorgi, Guiseppe, representation of the particle neighboring system.

By using the method for particle approximation, the properties of fluid dynamics are applied to particle i using the same neighboring system. This includes finding the value for density, pressure, and viscosity, which then can utilize the Navier-Stokes equation to find the acceleration of particle i (Schuermann). The density of this particle is found using the following equation.

$$\rho_i = \rho(A_i)$$

$$\rho_i = \rho_j \sum_{j=1}^{N} \frac{m_j}{\rho_j} w_{ij}$$

$$\rho_i = \sum_{j=1}^{N} \frac{m_j \rho_j}{\rho_j} w_{ij}$$

$$\rho_i = \sum_{j=1}^{N} m_j w_{ij}$$

Next is finding the pressure of particle i by using this equation.

$$p_i = -\nabla p(A_i)$$

$$p_i = -\nabla p_j \sum_{j=1}^{N} \frac{m_j}{\rho_j} w_{ij}$$

$$p_i = -\sum_{j=1}^{N} \frac{m_j p_j}{\rho_j} \nabla w_{ij}$$

Then finally to calculate the viscosity of particle i.

$$\mu_i = \mu \nabla^2 v(A_i)$$

$$\mu_i = \mu v \sum_{j=1}^{N} \frac{m_j \mu v}{\rho_j} \nabla^2 w_{ij}$$

These values are then applied to the particle within the code of the SPH simulator so that the particle's behavior, movement, and interactions are accurate to the properties of fluid

dynamics. Each particle in the simulation goes through these calculations, which then results in simulations with realistic water behavior. Due to having the same properties involved with water particles in real life, fluid simulation for testing real-world problems and models is very effective. This allows for an accurate demonstration of how the real model will function and interact with the fluids and environment involved.

Experiment

Within this paper, the use of fluid simulation will be experimented on different fundamental designs for hydraulic turbines. Computer software is used to run through these simulated scenarios, utilizing the SPH method for the representation of water and its interactions with objects. In this case, the shaft that would typically lead into a generator to create the power, is set in place by hinge points for the matter of simplicity of the demonstration. The software allows for many parameters to be manually tested for different fluids and designs such as velocity, density, viscosity, etc.

A few parameters were set in place for the results to be more consistent from each design and were done by establishing initial values and constants. The initial velocity from which the water is emitted is set at 2 meters per second. The density that will be used in this case is $\frac{1000kg}{m^3}$ as that is the density of freshwater.

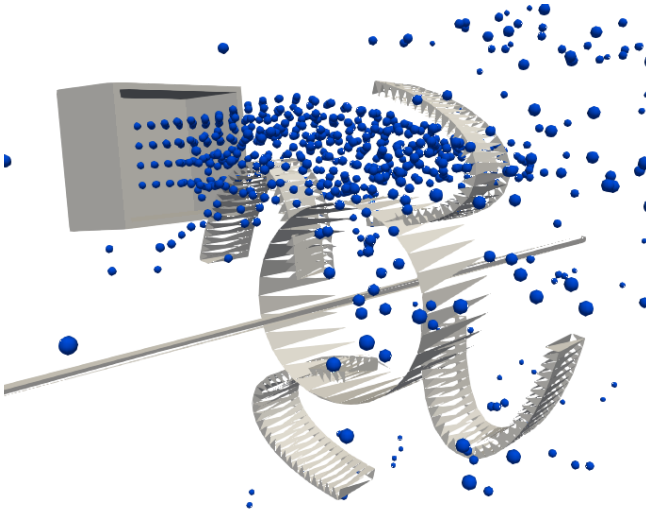


Figure 4. Simulation of impulse turbine.

Figure 4 is a model of an impulse turbine with a jet emitting water on the turbine. In this simulation, the water is being emitted into the runners to create rotational force, which in result turns the shaft. The force of water released by the jet is directly correlated with the amount of rotational force of the shaft. Simulating this scenario gives crucial information in finding the total power output as well as how each component of the design affects the rotation of the shaft. For example, the position of the jet affects the RPM of the shaft.

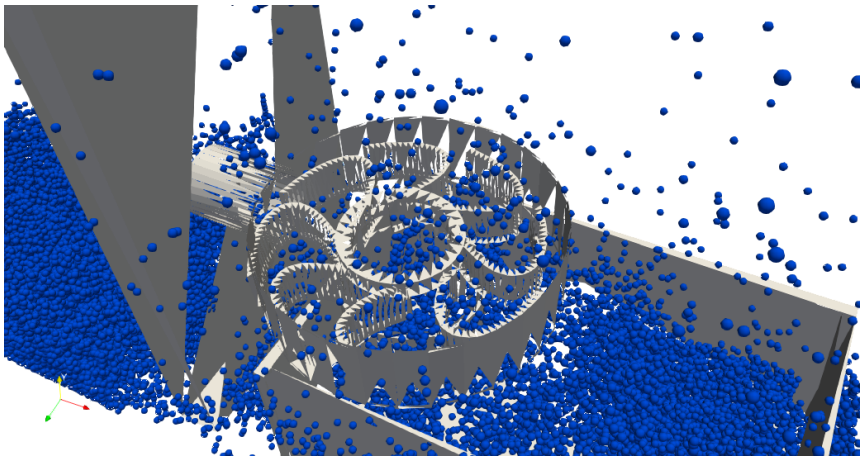


Figure 5. Simulation of reaction turbine.

Figure 5 is a model of a reaction turbine in which it has water being funneled in through a pipe. As the water is reflected off of the runners, it then goes down through the pipe within the middle of the turbine. The difference in water levels from the entry and exit points of the turbine causes a change in pressure, which is what causes the turbine to rotate and create energy. Factors such as the heights of the water and the size of pipes are very important in this simulation.

Calculations

From the information obtained from the simulations of both turbine designs, the total power output is able to be calculated. When calculating the power output of a reaction turbine, the formula $P = \Delta pV\eta$ is used, with Δp being the change in pressure on the turbine, V being the volumetric flow rate, and η being the efficiency of the turbine (Gummer). The change in pressure can be found with the density, gravity, and the change in water heights by using Poiseuille's law, $\Delta p = \rho g \Delta h$ (Nave). The efficiency is determined by the velocity of the water from the entry and exit points of the runner (mechcontent).

Reaction Turbine

$$\Delta p = \rho g \Delta h$$

$$\Delta p = \frac{1000kg}{m^3} \times \frac{9.81m}{s^2} \times 4.3m$$

$$\Delta p = \frac{98100kg}{ms^2} = 42183pa$$

The change in height is found by measuring the distance of the head within the design. This shows the correlation that the head has to the power output as the change in pressure directly affects the power.

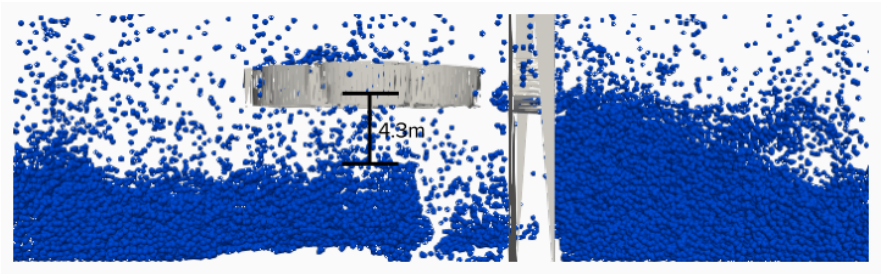


Figure 6. Representation of head.

When finding the volumetric flow rate, the mass flow rate must be found first. The mass flow rate is found with the density, cross-sectional area, and velocity, $\dot{m} = \rho A v$. The cross-sectional area is the area of the 2-dimensional shape that the water is flowing through, in this case, the circle of the pipe.

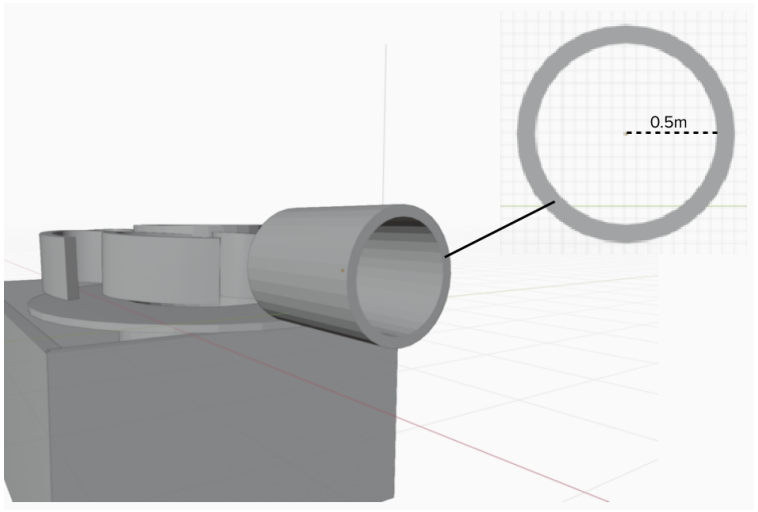


Figure 7. Determining cross-sectional area.

$$\dot{m} = \rho A v$$

$$\dot{m} = \frac{1000kg}{m^3} \times 0.25m^2 \times \frac{0.25\pi m}{s}$$

$$\dot{m} = \frac{196.35kg}{s}$$

The volumetric flow rate is the amount of volume that is moving per second and is calculated by dividing the mass flow rate and density.

$$V = \frac{\dot{m}}{\rho}$$

$$V = \frac{196.35kg}{s} \times \frac{m^3}{1000kg}$$

$$V = \frac{0.196m^3}{s}$$

The final piece to finding the power output is the efficiency of the blade. The efficiency of a runner is dependent on velocity values throughout the turbine. This is found by creating a velocity chart for the runner.

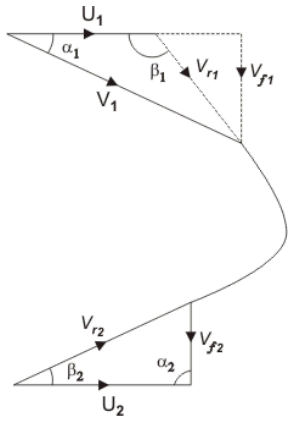


Figure 8. Chakraborty et al, Velocity chart of a runner on a reaction turbine.

The equation for finding efficiency is $\eta = \frac{2V_{f1}^2 cot\boldsymbol{\alpha}_1 (cot\boldsymbol{\alpha}_1 + cot\boldsymbol{\beta}_1)}{V_{f2}^2 + 2V_{f1}^2 cot\boldsymbol{\alpha}_1 (cot\boldsymbol{\alpha}_1 + cot\boldsymbol{\beta}_1)}$ (Chakraborty et al).

Trigonometry is used to find these values needed from the velocities of water entering and exiting the runner and the angles of the runner.

$$P = \Delta pV\eta$$

$$P = 42183pa \times \frac{0.196m^3}{s} \times 36.78\%$$

$$P = \frac{pam^3}{s}$$

$$P = 3040.92W$$

The total output of power from the reaction turbine is $3040.92W$.

Impulse Turbine

Next is finding the power output of the impulse turbine. Variables needed for this calculation include the mass flow rate ṁ, the change in relative velocity on the x-axis $(\Delta v_x)_R$, and the velocity on the x-axis V_x (Academy).

$$P = \dot{m}(\Delta v_x)_R V_x$$

The mass flow rate calculation is the same as the reaction turbine test.

$$\dot{m} = \frac{500kg}{s}$$

In order to find the velocity on the x-axis, the rpm and diameter of the turbine are needed. The diameter of the turbine is 2 meters and has a rpm of 24 which is found from running the simulation and recording the number of full rotations.

$$v_x = \frac{\pi r D}{60}$$

$$v_x = \frac{\pi \times 24rpm \times 1m}{60s}$$

$$v_x \approx \frac{1.26m}{s}$$

The next step is to find the value for $(\Delta v_x)_R$, which is where the design of the buckets has an effect on the values. Two values are needed as it is the change in velocity, so first the initial velocity will be calculated. This can be done using the velocity of water from the jet and the velocity on the x-axis found previously.

$$(v_x)_{Rinit} = v - v_x$$

$$(v_x)_{Rinit} = \frac{2m}{s} - \frac{1.26m}{s}$$

$$(v_x)_{Rinit} = \frac{0.74m}{s}$$

The final velocity will be the velocity of the water after hitting the change in the angle of the bucket. The cosine of that angle is then used to calculate the final velocity.

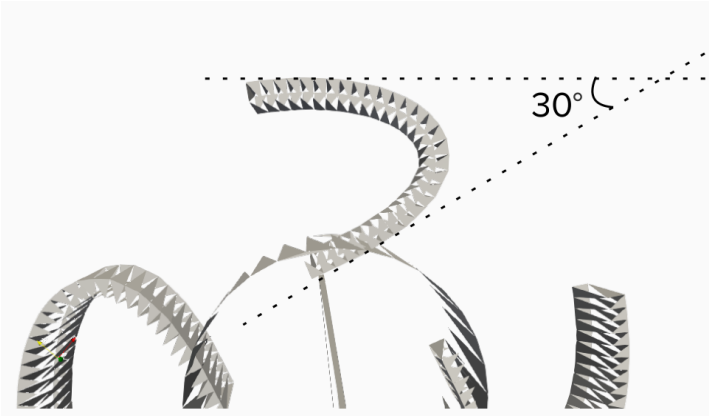


Figure 9. Angle of bucket.

$$(v_x)_{Rfinal} = (v_x)_{Rinit} \times cos(30°)$$

$$(v_x)_{Rfinal} = \frac{0.74m}{s} \times cos(30°)$$

$$(v_x)_{Rfinal} \approx \frac{0.64m}{s}$$

Hence, the change in velocity can be calculated. Due to the water's change in direction caused by the bucket, the velocities are added to calculate the change in velocity.

$$(\Delta v_x)_R = (v_x)_{Rinit} + (v_x)_{Rfinal}$$

$$(\Delta v_x)_R = \frac{0.74m}{s} + \frac{0.64m}{s}$$

$$(\Delta v_x)_R = \frac{1.38m}{s}$$

The total power output of the impulse turbine can now be calculated.

$$P = \dot{m}(\Delta v_x)_R v_x$$

$$P = \frac{500kg}{s} \times \frac{1.38m}{s} \times \frac{1.26m}{s}$$

$$P = \frac{711.9kgm^2}{s^3}$$

$$P = 869.4W$$

The total output of power from the reaction turbine is $869.4W$.

Conclusion

As seen from the results from both calculations, the reaction turbine made more power compared to the impulse turbine while being at a relatively similar scale. The reaction turbine would've produced 349.77% more power than the impulse turbine. A possible explanation for this is that the impulse turbine is receiving all the energy from the point at which the water jet is hitting, while the reaction turbine is submerged in water and is receiving energy from all points of the turbine (Woodford). Therefore, between these two simulations, the reaction turbine would be the most power-efficient turbine to build.

The introduction of CFD testing and production to turbine designs has left a huge impact on engineering. CFD is a very useful tool when it comes to design because it is able to test the designs before they are physically made. This process allows for money and resources to be saved by preventing the need to go through prototypes before the final product (ifes). It also allows for a much more extensive analysis of the design compared to experimental tests done in real life. It also provides a more time-efficient method in creating a design as it is able to be tested without it being built (aesmn). Another perk that comes with using a simulation is the configurability and ease of adjustment that it allows for and makes it very practical for real-life applications. For example, it allowed for the scale, position, and velocity to be adjusted for the water jet which can mimic the behavior of the jet being planned to use in real life. Along with that, it can also replicate possible obstacles in nature such as rocks and trees by creating objects within the simulation to visualize how those obstacles would affect the flow of the water.

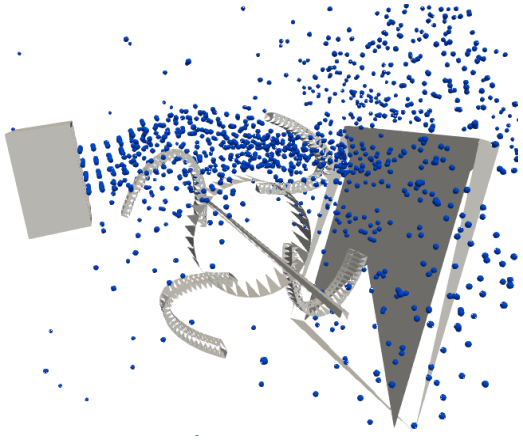


Figure 10. Impulse turbine model with the addition of an obstacle.

When it comes to making a more efficient design, the simulation provides an efficient and inexpensive method of testing alternative design options. For example, the angle used in the buckets for the impulse turbine impacts power output, which can be adjusted and compared to see what is the most optimal angle. If the turbine used salt water rather than freshwater, all that needs to be done is to adjust the values. For example, the density is different, so the value would change from 1000 to 1025 (Petrone).

There are limitations that come with this experiment as well as computational fluid dynamics as a whole. Within this experiment, the initial particles are emitted in a very uniform way and do not illustrate the movement of water until it interacts with other objects in the scene. Although this allows for consistent testing of each scene, it isn't realistic to how water would behave in the real world.

A limitation of the SPH method is that the number of particles is finite and therefore is set by the user. The number of particles can affect the results and cause inaccuracies through water not being represented appropriately from a lack of particles. The kernel function is also a limiting factor to SPH since some functions are more suitable depending on the conditions, materials involved, and fluids used (Zeng et al). Since it is performed on a computer, this

introduces limitations due to processing abilities, such as computation time. Certain designs due to complexity require a larger number of particles to obtain accurate results. If not enough computing values are present for the design, interpolation errors can occur (ifes). Hence, the more particles that are used within the simulation, the more calculations that are required by the computer, which increases the time it takes to run the simulation.

To reiterate, the use of fluid simulation has many benefits and is very effective within the design process of water turbines. Implementing this method of testing has the possibility of saving many people a lot of money and time. Along with that, it provides a more power-efficient turbine design, which in turn positively impacts the environment due to higher power output from clean energy. Therefore, fluid simulation should be more commonly used in creating designs.

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